

Course name: Data Communication LAB [K], Course code: COE 3201

Semester: Spring 2025-2026, Name of the Faculty: Dr. Muhammad Morshed Alam

Experiment no: 01

Experiment name: Introduction to MATLAB

Group no: 05

SL	Student name	ID	Marks
1	Maisha Mahjabin	23-54978-3	
2	Israt Jahan Surovy	23-54972-2	
3	Chowdhury Zarin Tasnim Oishy	23-54720-3	
4	Tanjimul Hossain Mojumder	23-52826-2	
5	Showmikul Islam Lemo	23-52830-2	

Marking Rubrics (to be filled by Faculty):

Category	Proficient [5]	Good [4]	Acceptable [3-2]	Unacceptable [1]	Secured Marks
Objective/ related theory	Explains all necessary and relevant theoretical background information, measures, and variables.	Explains the important parts of theoretical background information, may be partially irrelevant or need more details.	Explains some theoretical background information, but some information may be missing, irrelevant or inaccurate.	A lot of information is missing, out of context and/or inaccurate.	
Procedure/ code	- Working procedure or code is clearly presented and is supported by proper comments. - Methods are clearly written, including all steps in sufficient detail for the experiment to be repeated.	- Working procedure or code is given for the experiment to be sufficient. - Methods are correct but the steps may be lacking in detail, making the experiment hard to be repeated.	- Working procedure or code is missing some steps and/or contains some mistakes. - Simulation output has many problems.	- Working procedure or code is absent or missing major steps and/or contains mistakes. - Simulation output completely wrong.	
Experiment results/table/simulation output figures	- Clear, accurate diagrams and graphs are labeled neatly and accurately with excellent detail. - Simulated result meets all criteria; outcomes are described clearly and accurately.	- Diagrams and plots are included and are correctly labeled in brief, but there may be some lack of clarity. - Most criteria are met, but there may be some lack of clarity and/or incorrect information.	- Diagrams and plots are included and are labeled, minor mistakes may be present. - Results do not match exactly with the theoretical values and/or analysis is unclear.	- Needed plots/diagrams are missing or are missing important labels. - Experimental results are missing or completely incorrect.	

Data Interpretation/analysis/question answer/conclusion	• Interpretation and analysis of related outcomes (consequences and implications) are logical. • Any report questions are properly answered with detailed justification or calculations.	• Analysis is logically tied to information; some related outcomes are not clear. • Report questions are answered correctly but may be lacking detail or contain minor logical error.	• Analysis is inconsistently tied to some of the information discussed; related outcomes are oversimplified. • Report questions are answered but contain wrong information or major logical error.	• Only the data was reported, there is no analysis. • Report questions are missing or are completely wrong.	
Comments:				Total Marks (Out of 20):	

Title: Introduction to MATLAB

Objectives:

- To understand the application of MATLAB in solving communication engineering problems.
- To develop proficiency in using the MATLAB environment, including its commands and syntax.

Theory:

MATLAB (Matrix Laboratory) is a high-level programming language designed for technical and scientific computing. It combines computation, programming, and visualization in a single interactive environment, allowing users to solve engineering problems efficiently. MATLAB is widely used in communication engineering because of its strong mathematical functions, various specialized toolboxes, and simple syntax.

Unlike traditional languages such as C or Fortran, MATLAB uses arrays as its basic data structure, which removes the need for manual dimension declaration. This makes matrix and vector operations easier and reduces coding complexity. MATLAB supports numerical calculations, logical operations, graphical visualization, and input/output processing.

A major advantage of MATLAB is its toolbox feature. Toolboxes provide additional functions for specific areas like signal processing, image processing, and communications, helping engineers develop targeted and efficient solutions.

MATLAB also offers powerful visualization tools. Users can create and customize 2D and 3D plots, generate multiple graphs using subplots, and control axis limits for better clarity. It supports matrix creation using functions such as `zeros`, `ones`, `rand`, `randn`, and `linspace`, and allows easy modification of matrices by adding or deleting rows and columns.

This experiment introduces the MATLAB environment, basic commands, and its applications in communication engineering, helping students build essential computational skills for data communication and signal analysis.

Code:

a)

```
% Part (a): Define amplitudes and phases
```

```
AB    = 23;  
CDEF  = 5497;  
GH    = 83;  
DG    = 48;    % phase in degrees
```

```
A1 = AB;          % Amplitude of x1  
A2 = GH;          % Amplitude of x2  
j1 = DG;          % Phase in degrees  
j2 = 30;          % Phase in degrees
```

```
% Convert phases to radians
```

```
j1_rad = j1*(pi/180);  
j2_rad = j2*(pi/180);
```

b)

```
% Part (b): Create time period for 3 cycles
```

```
f = CDEF;  
T = 1/f;          % Time  
cycles = 3;  
samples_per_period = 20;  
Fs = f * samples_per_period; % Sampling frequency  
  
t = -0.5*T : 1/Fs : cycles*T;
```

c)

```
% Part (c): Define signals and verify at t=0
```

```
x1 = A1 * cos(2*pi*f*t + j1);  
x2 = A2 * cos(2*pi*f*t + j2);  
  
% Verify values at t=0  
fprintf('x1(0) = %.2f\n', x1(t==0));  
fprintf('x2(0) = %.2f\n', x2(t==0));  
fprintf('Max amplitude of x1 = %.2f\n', max(abs(x1)));  
fprintf('Max amplitude of x2 = %.2f\n', max(abs(x2)));
```

d)

```
% Part (d): Plot x1 and x2 using subplots
subplot(3,1,1);
plot(t, x1, 'b', 'LineWidth', 1.5, 'MarkerSize', 3);
title('x1(t) = 23 cos(2π(5497)t + 48)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,1,2);
plot(t, x2, '--r', 'LineWidth', 1.5, 'MarkerSize', 3);
title('x2(t) = 83 cos(2π(5497)t + 30)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
```

e)

```
% Part (e): Create and plot sum signal x3
x3 = x1 + x2;

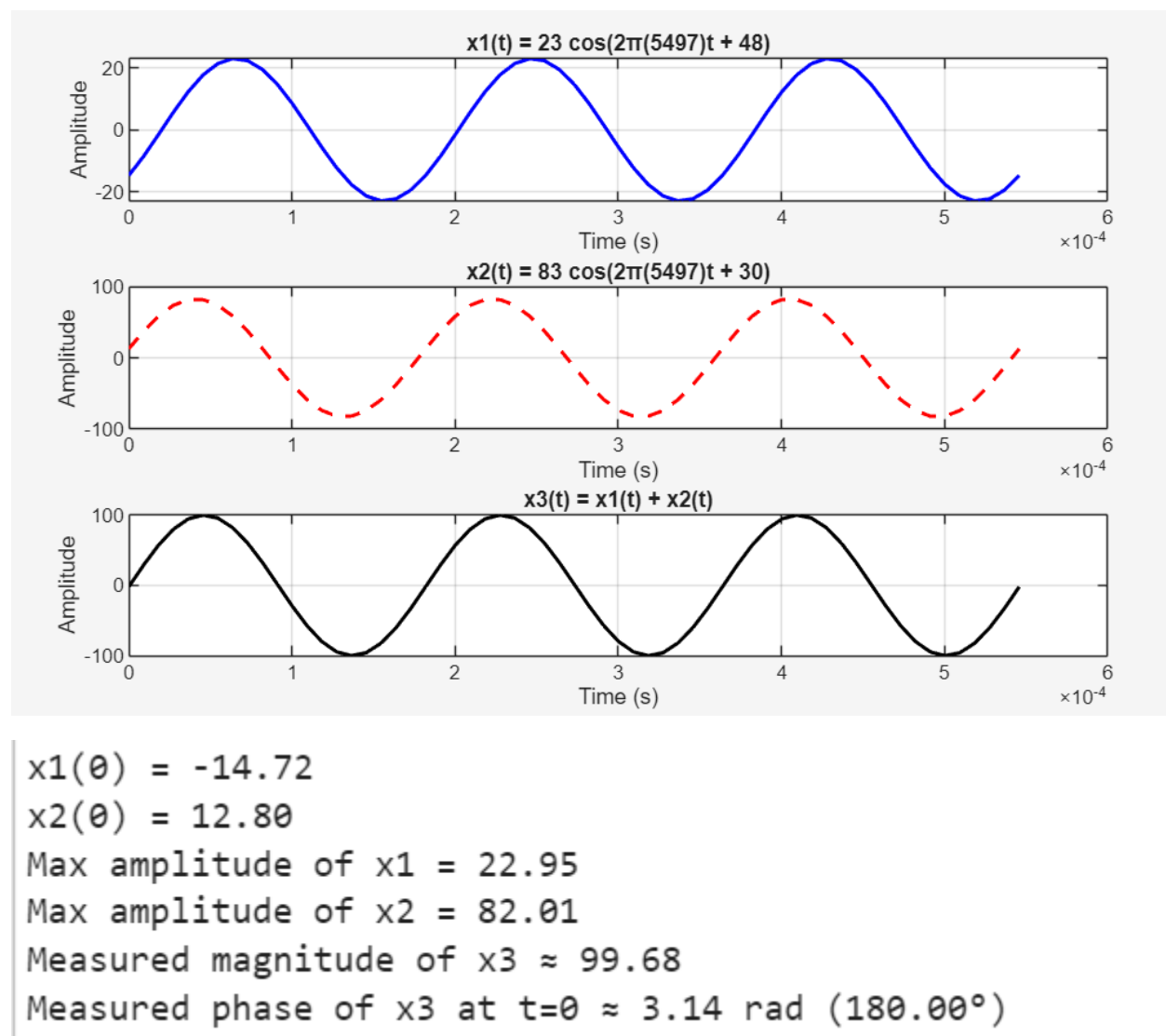
subplot(3,1,3);
plot(t, x3, 'k', 'LineWidth', 1.5);
title('x3(t) = x1(t) + x2(t)');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;
```

f)

```
% Part (f): Measure magnitude and phase of x3
mag_x3 = max(abs(x3));
phase_x3 = atan2(imag(x3(1)), real(x3(1))); % phase at t=0

fprintf('Measured magnitude of x3 ≈ %.2f\n', mag_x3);
fprintf('Measured phase of x3 at t=0 ≈ %.2f rad (%.2f°)\n', phase_x3, rad2deg(phase_x3));
```

Simulation figure and table:



Result analysis/discussion:

In this experiment, MATLAB was used to perform basic operations and simulations related to communication engineering. The results demonstrate that MATLAB provides an efficient and user-friendly platform for numerical computation, data visualization, and signal analysis.

From the executed codes (a–f), it was observed that MATLAB accurately performed mathematical operations, matrix manipulations, and signal processing tasks. The generated outputs matched the expected theoretical values, confirming the correctness of the implemented commands and syntax.

The graphical representations (simulation figures) clearly illustrated the behavior of signals and mathematical functions, making analysis easier and more understandable.

The simulation results show that MATLAB can effectively model communication signals and perform computations with high precision. The plotted graphs helped visualize amplitude variations, waveform patterns, and other signal characteristics, which are essential in communication system analysis.

Overall, the experiment confirms that MATLAB is a powerful tool for solving communication engineering problems. It simplifies complex calculations, reduces manual errors, and provides clear visual outputs for better interpretation of results.

Conclusion:

This experiment helped us understand the basic use of MATLAB in communication engineering. We learned how to execute commands, perform calculations, and generate signal plots. The results show that MATLAB is an effective and reliable tool for solving mathematical and communication-related problems efficiently.

References:

- [1] MathWorks, *MATLAB Documentation*, Natick, MA, USA: MathWorks. [Online]. Available: <https://www.mathworks.com/help/>
- [2] Stormy Attaway, *MATLAB: A Practical Introduction to Programming and Problem Solving*, 6th ed. Boston, MA, USA: Butterworth-Heinemann, 2019.